



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Assignment 2 - Semester 2, 2024

Assessment Title: Big Data Combo Kit

Assessment Outcome: 30%

Due Date: AEST 23:59 on 25/10.

Assessable Item:

- One (1) piece of a report containing the complete flowchart showing your design idea about preparing the data, the tools used in all steps, and the report for answers to the given questions.
- One (1) piece of a doc or pdf format code you wrote for the assignment. Please note that using pictures or screenshots is not allowed. Only format that can be examined by Turnitin is acceptable.

Purpose of Assignment

This assignment tests whether a student is capable of using all major tools learned in this unit to cope with real-world problems and achieve the specific given goal. The design should be reasonable, practical, and manageable. You must use a single execution of the designed tool to answer all questions in this part. Using different designs to answer independent tasks will result in significant marks loss.

Part 1: Finding Corresponding Information from Multiple Sources

Introduction

You are given a task to use algorithms, tools, and skills taught in this unit to identify the locations of specific sentences in the sources. In your virtual machine, there are two text sources named "ratings_2012.txt" and "ratings_2013.txt" located under the path "/home/training/training_materials/analyst/data." Make your design and combination of tools to answer the questions in the assignment task.

Assignment Task

The following tasks should be accomplished:

- 1.1. Explain how you manage the tool you chose to access multiple files in one go. **[5 marks]**
- 1.2. Provide the full source code you used for this assignment. The source code must match the design you used for the first task. **[20 marks]**
- 1.3. Show the screenshot of all locations in your result where the sentence “*Item was defective*” appears in the sources. **[10 marks]**
- 1.4. Show the screenshot of all locations in your result where the comment with only a single word, “*Shoddy*”, appears in the sources. **[15 marks]**

Submission Requirement

You must use the template to prepare your assignment. For the source code, you need to copy the codes and put it in the text file(s), the word file(s), or the pdf file(s) and submit it(them) to Turnitin. You can only use **one** design (tool) **in a single run** to complete **all** tasks in this part.

Part 2: RDBMS and HiveQL

Introduction

Your mission in this assignment is to process the raw data from the text file, align the data to the format that matches the standard in the relational database, and put the data into the HDFS Hive for information retrieval. The collected data is assumed to have no misleading information. The linguists generally use special characters in this field. You don't need to do data cleaning in this project, e.g., leave the contents untouched but arrange them into the proper format before putting them into Hive or MySQL.

The given dataset is a manageable size, but we assume it is an extensive dataset; thus, you are allowed to use only the tools introduced in this unit. It means that you should properly combine the usage of MapReduce, Pig Latin, and HiveQL to complete the tasks in this project. Using other tools for handling regular data size will NOT be considered a feasible solution in the assessment. Even if the result is correct, you will lose marks if the results are not from eligible tools.

The column names of the given dataset are listed below, along with the first two columns of the first tuple in the data (see Table 1).

Table 1. Attribute headers and the sample tuple.

language_code	language_name	language_synonym	thesaurus_heading_language	thesaurus_heading_people	approximate_latitude_of_language_variety	approximate_longitude_of_language_variety	state_territory	uri
A1	Nyaki Nyaki / Njaki Njaki							

Assume the language_code is unique. A valid relational database (RDB) record for this tuple should be split into two pairs: (A1, Nyaki Nyaki) and (A1, Njaki Njaki), where “A1” serves as the key and the “Nyaki Nyaki” and “Njaki Njaki” are corresponding values.

The remaining attributes can also be decomposed by the same method. To create an efficient RDB, you should have multiple tables to contain the necessary data and use the weak entity to capture the relations between tables. In the example above, you should decompose the whole table into

two tables: language_code and language_name. The language_code table would contain the code “A1”, and the language_name table would have two tuples obtained from the given example tuple: “Nyaki Nyaki” and “Njaki Njaki”. The weak entity is the third table to capture the pairs (A1, Nyaki Nyaki) and (A1, Njaki Njaki). The same concept can be extended to other attributes as well. A reasonable designed ERD for decomposing the given dataset is revealed below. The first ERD is for the MySQL RDBMS (see Fig. 1) and the second ERD is for HiveQL (see Fig. 2).

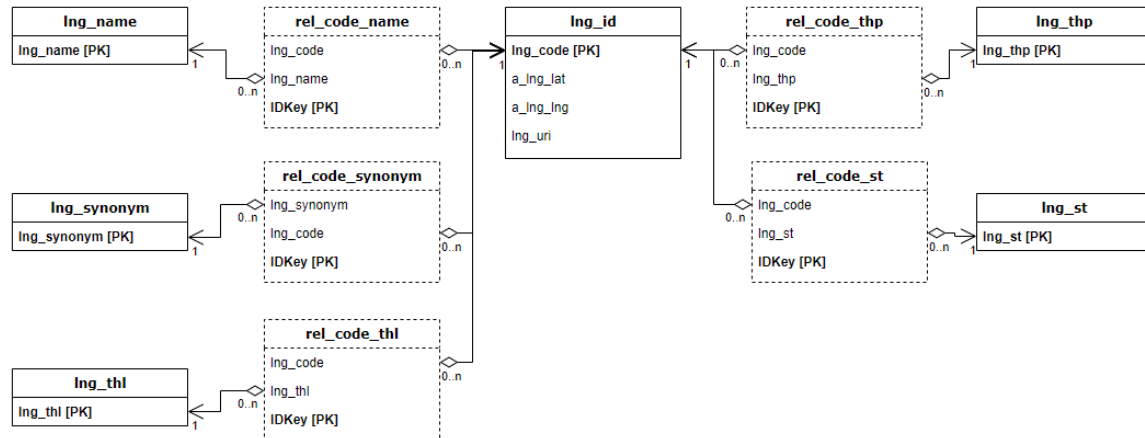


Fig. 1. ERD for MySQL

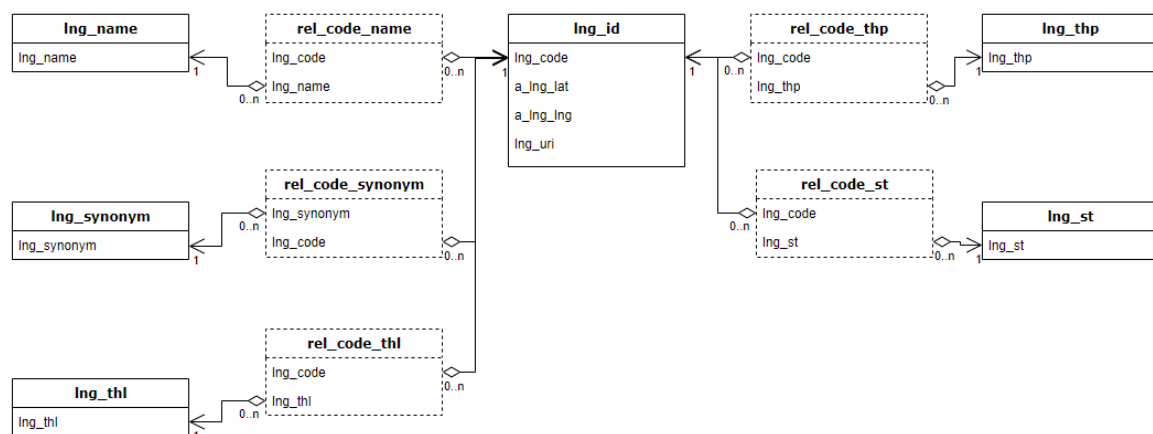


Fig. 2. ERD for HiveQL

The attribute name mapping in the original dataset to the ERDs is listed in Table 2. The first row indicates the attribute name used in the source file, and the second row is the new name used in the ERD.

Table 2. Mapping of the attribute names in the source file and the ERDs

language_code	language_name	language_synonym	thesaurus_heading_language	thesaurus_heading_people	approximate_latitude_of_language_variety	approximate_longitude_of_language_variety	state_territory	uri
lng_code	lng_name	lng_synonym	lng_thl	lng_thp	a_lng_lat	a_lng_lng	lng_st	lng_uri

You are expected to choose wisely the tool for processing the data, convert the source data into something that exactly matches the given ERDs, and then use query statements to find answers to the tasks. The total length of your report should be no more than 20 pages. Anything beyond the 20th page will not be marked. You should apply the training you had in the past to organise the

content efficiently to communicate your answer to the reader. This time, there will be no guide on what to put in the solution for a specific task. You can find clues from past assignments to clearly organise your report and express the concept.

Assignment Goal

This assignment aims to train students to analyse the encountered problems and find the most suitable tool(s) for accomplishing the given tasks in the real-world big data processing environment. Students will encounter some research and discover components to learn skills and knowledge from the project.

Assignment Task

The following tasks should be accomplished:

- 2.1. Find a way with tools taught in this unit to check whether the language_code attribute in the source data is unique. **Note that you cannot eyeball the result. The result should be found by combining valid tools in a sequence.**
- 2.2. Find a way with tools taught in this unit to list the unique values for all entities (tables with the solid boundary) except lng_id.
- 2.3. Find a way with tools taught in this unit to prepare the data for all entities (tables with the solid boundary) for MySQL and Hive.
- 2.4. Find a way with tools taught in this unit to prepare the data for all weak entities (tables with the dashed boundary) for MySQL and Hive.
- 2.5. Follow the given ERDs, create tables and import data into the created tables. All tables should be put in the database entitled "indigenous".
- 2.6. Show the detailed information of table rel_code_st in MySQL.
- 2.7. Collect the results containing "lng_code", "lng_name", and "lng_st" of tuples whose lng_name starts with the upper case "D". Show both statements and the last page screenshot of your query result in MySQL and Hive.
- 2.8. Collect the results containing "lng_code", "lng_name", "lng_st", "a_lng_lat", and "a_lng_lng" of tuples whose lng_synonym contains "Kerama". Show both statements and the last page screenshot of your query result in MySQL and Hive.

Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template is acceptable. Your submission will be marked only if your submitted code passes the examination.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Assignment 2

	Task	Mark
1.1	Explanation of your design.	5 marks
1.2	Clear and neat source code with detailed comments.	20 marks

1.3	Screenshot of the output.	10 marks
1.4	Screenshot of the output.	15 marks
2.1	Correctness of the answers.	1.5 marks for each subtask. (37.5 marks in total)
2.2	The efficiency of the proposed solutions.	1 marks each for task 1-4, 1.5 marks for task 7, and 2 marks for task 8. (7.5 marks in total)
2.3	The report content is explicit, neat, and easy to understand.	- 3.7 to 5 marks: Professional writing, is explicit, neat, easy to understand and contains no mistakes. - 2.6 to 3.6 marks: Outstanding writing, neat, easy to understand and contains none or tinny mistakes. - 1.3 to 2.5 marks: Average writing; some mistakes can be found in the answers. - 0 to 1.2 marks: General writing. Missing major components to piece back the designed method.

----- End of Assignment -----